Similar areas and volumes (Extended)

Core Concepts: Similar areas and volumes (Extended)

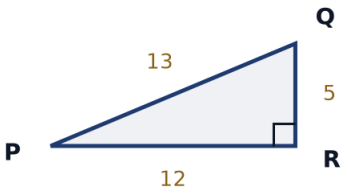
1 In $\triangle ABC$ studying similar areas and volumes (extended), $m\angle A = 36^\circ$ and $m\angle B = 45^\circ$. What is $m\angle C$?

a. 99°

b. 109°

c. 91°

d. 144°



2 A geometric figure scaled by factor $k = 2$ has an original side length of 5 cm. What is the corresponding image length?

a. 8 cm

b. 10 cm

c. 15 cm

d. 20 cm

3 On the coordinate plane, a line with slope $m = 4$ passes through $(0, 4)$. What is its y-coordinate when $x = 1$?

a. 4

b. 10

c. 8

d. 12

Procedural Fluency & Algebraic Manipulation

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4 In $\triangle ABC$ studying similar areas and volumes (extended), $m\angle A = 36^\circ$ and $m\angle B = 52^\circ$. What is $m\angle C$?

a. 92°

b. 102°

c. 84°

d. 144°

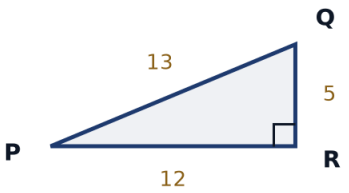
5 A geometric figure scaled by factor $k = 3$ has an original side length of 5 cm. What is the corresponding image length?

a. 12 cm

b. 15 cm

c. 20 cm

d. 45 cm



6 On the coordinate plane, a line with slope $m = 4$ passes through $(0, 4)$. What is its y-coordinate when $x = 2$?

a. 8

b. 14

c. 12

d. 16

Graphical & Geometric Interpretation

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7 In $\triangle ABC$ studying similar areas and volumes (extended), $m\angle A = 36^\circ$ and $m\angle B = 59^\circ$. What is $m\angle C$?

a. 85°

b. 95°

c. 77°

d. 144°

8 A geometric figure scaled by factor $k = 4$ has an original side length of 5 cm. What is the corresponding image length?

a. 16 cm

b. 20 cm

c. 25 cm

d. 80 cm

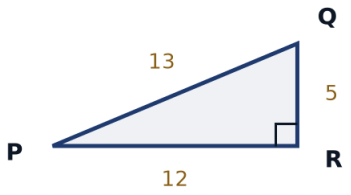
9 On the coordinate plane, a line with slope $m = 4$ passes through $(0, 4)$. What is its y-coordinate when $x = 3$?

a. 12

b. 18

c. 16

d. 20



Exam-Style Problem Solving & Modeling

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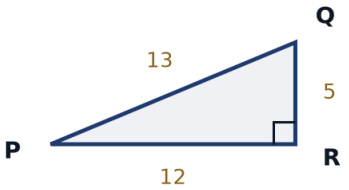
10 In $\triangle ABC$ studying similar areas and volumes (extended), $m\angle A = 36^\circ$ and $m\angle B = 66^\circ$. What is $m\angle C$?

a. 78°

b. 88°

c. 70°

d. 144°



11 A geometric figure scaled by factor $k = 5$ has an original side length of 5 cm. What is the corresponding image length?

a. 20 cm

b. 25 cm

c. 30 cm

d. 125 cm

12 On the coordinate plane, a line with slope $m = 4$ passes through $(0, 4)$. What is its y-coordinate when $x = 4$?

a. 16

b. 22

c. 20

d. 24

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